

Week 2 Progress

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July 2026

This is a team project

Hogan

- Weather and heat framing
- Goggins baseline challenge (Jan)
- Ages 65–69 and 70–74
- Climate X file and hot-month definitions

Roro

- Hospital Authority stroke aggregates
- Outcome timing and transfer path

Dr Bishai

- Original multi-method plan
- Teamwork: go far together

Also in the room this week

- Writing early (lit + methods) matters
- Equity lens (Marmot): who bears risk
- Peer heat work: summit, district voice, LegCo

This update credits the team's contributions. Judgments on weather and outcomes are still shared.

What the July meeting changed

Earlier plan	Meeting	Working now
AMI + stroke principal-dx	No admission reasons in general HA file	Stroke aggregates; AMI out of scope from that file
One primary heat model	Explore many methods	Labelled multi-spec panel
Heat metrics	Align with Ren/Wang-style spells	Spells, 2D3N, hot-month options with Hogan

Roro's outcome path (working understanding): first GOPC mention of stroke is a *marker*. The actual stroke and hospitalisation generally come earlier. We ignore later mentions and work back to the true event month.

The question we are preparing

How do **monthly** thermal exposures in Hong Kong, 2013–2023, associate with **stroke event aggregates**, and how stable is that picture across a shared panel of heat, cold, spell, and sensitivity specifications?

Complementary to daily heatwave–mortality work in the lab — not a duplicate. Not a daily DLNM; not AMI from a file without admission reasons.

From January: Hogan's intellectual lead

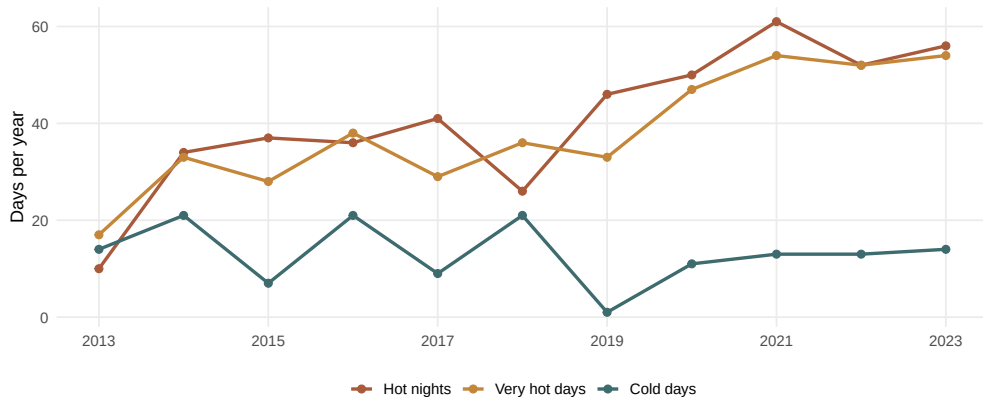
- Pulled Goggins MI (2013) and stroke (2012) as the Hong Kong baseline
- Warned that cold associations are already well documented — novelty must be careful
- Suggested comparing a later period (2013–2023) as an evolutionary question
- Highlighted ages **65–69** and **70–74** as actionable older bands
- Offered to prepare the monthly climate *X* file under Dr Bishai's plan
- Later: define heatwaves, count them by month, flag upper-tail “hot” months (Atmos. Res. heatwave framing; DOI 10.1016/j.atmosres.2024.107845)

Those contributions still structure the weather side. Definitions should be finished **with** Hogan, not around him.

Climate: shared ground truth (descriptive)

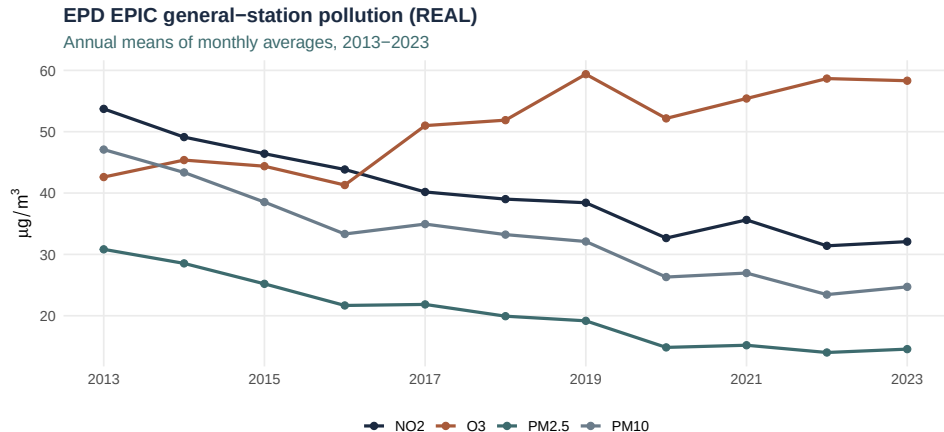
Official thermal extremes at HKO Headquarters

Annual counts, 2013–2023 (environmental data only)



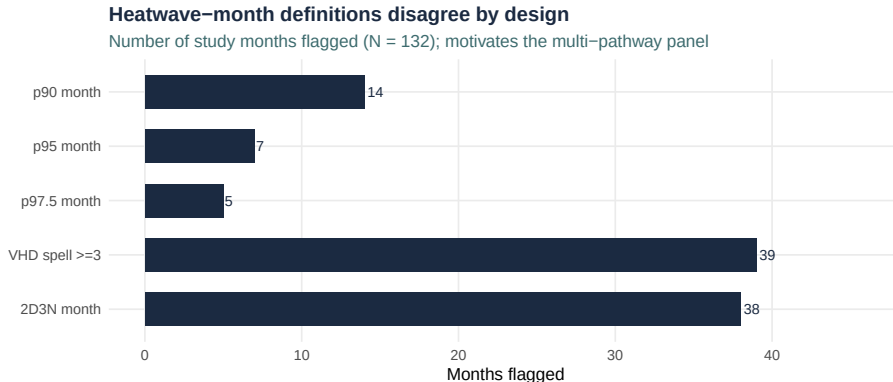
Hot nights and very hot days rose after 2018; cold days persist. Peak hot-night year in-window: **2021 (61)**. Environmental descriptives only — no stroke findings.

Pollution layer (EPD EPIC)



Official monthly general-station means: NO₂, O₃, PM_{2.5}, PM₁₀. NO₂/PM declined; O₃ rose — echoing the January discussion of a changing pollutant mix. Stage pollutants carefully; enter O₃ last.

Why heatwave definitions stay plural



Guo/Gasparrini, Wang/Ren, and the lab's multi-definition mortality work all argue against one universal rule. **Hogan's hot-month idea** (heatwave counts $\rightarrow$ upper-tail months) belongs here as a named option.

A labelled panel — not seventeen discoveries

Family	Examples (lit-anchored)
Continuous T / lag	Tmean; Tmax/Tmin; lag-1
Official extremes	Hot nights, very hot days, cold days (Hogan / HKO)
Spells & 2D3N	Ren/Wang-style burden metrics
Hot-month defs	Percentile months + Hogan count→tail idea
Cold & flu	Cold-primary; influenza co-exposure
Structure	Age bands Hogan flagged; sex if grain allows
Sensitivities	Pollution staging; humidity; COVID window

Headline proposal for Gate 3 *with the team*: continuous Tmax/Tmin + official extreme-day counts.
Full pathway×literature map is in the Laidlaw write-up.

What each person still unlocks

Week 2 readiness: one missing piece

Public layers and analysis plumbing are ready; stroke aggregates unlock inference



Hogan: final weather / heatwave / hot-month operationalisation.

Roro: stroke aggregates — first GOPC mention → true event month; dictionary; transfer.
Public pollution and population layers can support the merge once outcomes arrive.

Honesty line

Ready to discuss

- Environmental descriptives
- Outcome construction logic with Roro
- Weather definitions with Hogan
- Lit + methods writing for Laidlaw

Not ready to claim

- Stroke coefficients
- “Heat replaced cold”
- AMI from this HA file
- Synthetic dry-runs as findings

Literature anchors (selected)

- **Goggins 2012/2013** — cold-dominant HK stroke/AMI baselines (Hogan's challenge)
- **Guo 2024** — hot-night intensity vs official binary flags
- **Wang/Ren 2019** — VHD/HN spells and 2D3N
- **Guo/Gasparrini 2017** — multi-definition heatwaves
- **Yang/Chong 2025** — cold + influenza and stroke
- **Guo 2025** — pollution can amplify temperature–hospitalisation risks

Laidlaw Stage 3 is on the horizon

- Research essay (2000–3000 words) — lit + methods drafts are started
- A0 **portrait** poster (841×1189 mm) — research focus; results pending outcomes
- HKU report form + supervisor endorsement
- Showcase: HKU Laidlaw Society (early Nov 2026) and Annual Conference

Writing early — as other presenters modelled — is how we stay honest about what is ready and what still needs the team.

Thank you

Next small asks — whenever you have a moment :)

- **Hogan:** shall we lock the heatwave / hot-month definitions together (including your count→upper-tail idea)?
- **Roro:** when you can, a short note on the first-GOPC→true-stroke timing and the aggregate file path would unlock the merge.

If you want to go fast, go alone; if you want to go far, go together.

Grateful for both of you — and for Dr Bishai's nudge to listen better.